

Exercises for Relational Algebra

Use the COMPANY database from Query 1 to Query 7.

Query 1. Retrieve the name and address of all employees who work for the 'Research' department.

```
RESEARCH_DEPT  $\leftarrow \sigma_{Dname='Research'}(DEPARTMENT)$   
RESEARCH_EMPS  $\leftarrow (RESEARCH\_DEPT \bowtie_{Dnumber=Dno} EMPLOYEE)$   
RESULT  $\leftarrow \pi_{Fname, Lname, Address}(RESEARCH\_EMPS)$ 
```

As a single in-line expression, this query becomes:

```
 $\pi_{Fname, Lname, Address}(\sigma_{Dname='Research'}(DEPARTMENT \bowtie_{Dnumber=Dno}(EMPLOYEE)))$ 
```

Query 2. For every project located in 'Stafford', list the project number, the controlling department number, and the department manager's last name, address, and birth date.

```
STAFFORD_PROJS  $\leftarrow \sigma_{Plocation='Stafford'}(PROJECT)$   
CONTR_DEPTS  $\leftarrow (STAFFORD\_PROJS \bowtie_{Dnum=Dnumber} DEPARTMENT)$   
PROJ_DEPT_MGRS  $\leftarrow (CONTR\_DEPTS \bowtie_{Mgr\_ssn=Ssn} EMPLOYEE)$   
RESULT  $\leftarrow \pi_{Pnumber, Dnum, Lname, Address, Bdate}(PROJ\_DEPT\_MGRS)$ 
```

Query 3. Find the names of employees who work on all the projects controlled by department number 5.

$$\begin{aligned} \text{DEPT5_PROJS}(\text{PNO}) &\leftarrow \pi_{\text{PNUMBER}}(\sigma_{\text{DNUM}=5}(\text{PROJECT})) \\ \text{EMP_PROJ}(\text{SSN}, \text{PNO}) &\leftarrow \pi_{\text{ESSN}, \text{PNO}}(\text{WORKS_ON}) \\ \text{RESULT_EMP_SSNS} &\leftarrow \text{EMP_PROJ} \div \text{DEPT5_PROJS} \\ \text{RESULT} &\leftarrow \pi_{\text{LNAME}, \text{FNAME}}(\text{RESULT_EMP_SSNS} * \text{EMPLOYEE}) \end{aligned}$$

Query 4. Make a list of project numbers for projects that involve an employee whose last name is 'Smith', either as a worker or as a manager of the department that controls the project.

$$\begin{aligned} \text{SMITHS}(\text{Essn}) &\leftarrow \pi_{\text{Ssn}}(\sigma_{\text{Lname}='Smith'}(\text{EMPLOYEE})) \\ \text{SMITH_WORKER_PROJS} &\leftarrow \pi_{\text{Pno}}(\text{WORKS_ON} * \text{SMITHS}) \\ \text{MGRS} &\leftarrow \pi_{\text{Lname}, \text{Dnumber}}(\text{EMPLOYEE} \bowtie_{\text{Ssn}=\text{Mgr_ssn}} \text{DEPARTMENT}) \\ \text{SMITH_MANAGED_DEPTS}(\text{Dnum}) &\leftarrow \pi_{\text{Dnumber}}(\sigma_{\text{Lname}='Smith'}(\text{MGRS})) \\ \text{SMITH_MGR_PROJS}(\text{Pno}) &\leftarrow \pi_{\text{Pnumber}}(\text{SMITH_MANAGED_DEPTS} * \text{PROJECT}) \\ \text{RESULT} &\leftarrow (\text{SMITH_WORKER_PROJS} \cup \text{SMITH_MGR_PROJS}) \\ &\pi_{\text{Pno}}(\text{WORKS_ON} \bowtie_{\text{Essn}=\text{Ssn}} (\pi_{\text{Ssn}}(\sigma_{\text{Lname}='Smith'}(\text{EMPLOYEE}))) \cup \pi_{\text{Pno}} \\ &((\pi_{\text{Dnumber}}(\sigma_{\text{Lname}='Smith'}(\pi_{\text{Lname}, \text{Dnumber}}(\text{EMPLOYEE}))) \bowtie_{\text{Ssn}=\text{Mgr_ssn}} \text{DEPARTMENT})) \bowtie_{\text{Dnumber}=\text{Dnum}} \text{PROJECT}) \end{aligned}$$

Query 5. List the names of all employees with two or more dependents.

```
T1(Ssn, No_of_dependents) ←  $\pi_{\text{Essn}} \bowtie \text{COUNT } \text{Dependent\_name}(\text{DEPENDENT})$   
T2 ←  $\sigma_{\text{No\_of\_dependents} > 2}(T1)$   
RESULT ←  $\pi_{\text{Lname, Fname}}(T2 * \text{EMPLOYEE})$ 
```

Query 6. Retrieve the names of employees who have no dependents.

```
ALL_EMPS ←  $\pi_{\text{Ssn}}(\text{EMPLOYEE})$   
EMPS_WITH_DEPS(Ssn) ←  $\pi_{\text{Essn}}(\text{DEPENDENT})$   
EMPS_WITHOUT_DEPS ← (ALL_EMPS – EMPS_WITH_DEPS)  
RESULT ←  $\pi_{\text{Lname, Fname}}(\text{EMPS\_WITHOUT\_DEPS} * \text{EMPLOYEE})$ 
```

Query 7. List the names of managers who have at least one dependent.

```
MGRS(Ssn) ←  $\pi_{\text{Mgr\_ssn}}(\text{DEPARTMENT})$   
EMPS_WITH_DEPS(Ssn) ←  $\pi_{\text{Essn}}(\text{DEPENDENT})$   
MGRS_WITH_DEPS ← (MGRS  $\cap$  EMPS_WITH_DEPS)  
RESULT ←  $\pi_{\text{Lname, Fname}}(\text{MGRS\_WITH\_DEPS} * \text{EMPLOYEE})$ 
```

Use the RESTAURANT database from Query 8 to Query 16..

<i>Person</i> (name, age, gender)	name is a primary key
<i>Frequents</i> (name, pizzeria)	(name, pizzeria) is a composite primary key
<i>Eats</i> (name, pizza)	(name, pizza) is a composite primary key
<i>Serves</i> (pizzeria, pizza, price)	(pizzeria, pizza) is a composite primary key

Query 8. Find all pizzerias frequented by at least one person under the age of 18.

Query 9. Find the names of all females who eat either mushroom or pepperoni pizza (or both).

Query 10. Find the names of all females who eat both mushroom and pepperoni pizza.

Query 11. Find all pizzerias that serve at least one pizza that Amy eats for less than \$10.00.

Query 12. Find all pizzerias that are frequented by only females or only males.

Query 13. For each person, find all pizzas the person eats that are not served by any pizzeria the person frequents. Return all such person (name) / pizza pairs.

Query 14. Find the names of all people who frequent only pizzerias serving at least one pizza they eat.

Query 15. Find the names of all people who frequent every pizzeria serving at least one pizza they eat.

Query 16. Find the pizzeria serving the cheapest pepperoni pizza. In the case of ties, return all of the cheapest-pepperoni pizzerias.

Query 8: $\pi_{pizzeria}(\sigma_{age < 18}(Person) * Frequent)$

Query 9: $\pi_{name}(\sigma_{gender='female' \wedge (pizza='mushroom' \vee pizza='pepperoni')}(Person * Eats))$

Query 10: $\left(\pi_{name}(\sigma_{gender='female' \wedge pizza='mushroom'}(Person * Eats)) \right) \cap$
 $\left(\pi_{name}(\sigma_{gender='female' \wedge pizza='pepperoni'}(Person * Eats)) \right)$

Query 11: $\pi_{pizzeria}((\sigma_{name='Amy'}(Eats)) * (\sigma_{price < 10}(Serves)))$

Query 12: $\left(\left(\pi_{pizzeria}(\sigma_{gender='female'}(Person) * Frequent) \right) - \left(\pi_{pizzeria}(\sigma_{gender='male'}(Person) * Frequent) \right) \right) \cup$
 $\left(\left(\pi_{pizzeria}(\sigma_{gender='male'}(Person) * Frequent) \right) - \left(\pi_{pizzeria}(\sigma_{gender='female'}(Person) * Frequent) \right) \right)$

Query 13: $Eats - (\pi_{name,pizza}(Frequent * Serves))$

Query 14: $\pi_{name}(Person) - (\pi_{name}(Frequent - \pi_{name,pizzeria}(Eats * Serves)))$

Query 15: $\pi_{name}(Person) - (\pi_{name}(\pi_{name,pizzeria}(Eats * Serves) - Frequent))$

Query 16: $\pi_{pizzeria}(\sigma_{pizza='pepperoni'} Serves) -$
 $\pi_{pizzeria} \left(\sigma_{price > price2} \left(\left(\pi_{pizzeria,price}(\sigma_{pizza='pepperoni'} Serves) \right) * \left(\rho_{pizzeria2,price2}(\pi_{pizzeria,price}(\sigma_{pizza='pepperoni'} Serves)) \right) \right) \right)$